General

In Inspera, questions 1-6 have 10 points. This must be ignored. We use 0-3 points for every question. As the exam cover page states, every question has the same value for the exam.

Topic OS1: virtual memory

	Question in English	**Spørsmål på norsk**	**Retteveiledning (hvor mange poeng gis)**
	Operating systems can only implement virtual memory if the computer they run on provides hardware support for this. All major desktop OSes require that this is available.	Operativsystemer kan bare implementere virtuelt minne hvis datamaskinen de kjører på gir maskinvarestøtte for dette. Alle større stasjonære operativsystemer krever at dette er tilgjengelig.	This is not a question.
1	Explain how a memory lookup works in a computer that uses a single-level page table to implement virtual memory (conceptually, not specifically for one processor). Distinguish the two cases where the page is available in memory and where it is available from secondary storage.	Forklar hvordan et minneoppslag fungerer på en datamaskin som bruker en sidetabell på ett nivå for å implementere virtuelt minne (konseptuelt, ikke spesifikt for én prosessor). Skill de to tilfellene der siden er tilgjengelig i minnet og der den er tilgjengelig fra sekundær lagring.	• 3: Processor receives an instruction to read from memory, passes the address to the MMU. Only the first bits of the address are required for lookup (the rest is offset). The MMU looks this prefix up in the page table. There is a present bit for the page. If it is set, we have case 1 and the entry contains the physical address. The offset is added and it is returned to the processor. If it is not set, we have case 2. A page fault is generated, the OS must (optional: probably move another page to swap space, and) load the missing page into physical memory, update the page table, set the present bit, and add the offset to the new base address, and restart the lookup procedure. • 2: Up to two of the points above are missing

			and the rest is good, or everything is present but inaccurately described. ● 1: A virtual address must be translated to a physical address. There is a thing called a page table, which contains a translation if the page it present. Some indication that something must happen if it is not present. (understand the question and provide a little bit more info)
2	The fork() instruction creates a new process. Immediately after fork(), the old and new processes appear identical apart from their process IDs. State the main challenge of this behavior and how virtual memory makes it possible. Explain the role of the page tables.	Fork()-instruksjonen oppretter en ny prosess. Umiddelbart etter fork(), ser den gamle og den nye prosessen identiske ut bortsett fra prosess-ID-ene. Angi hovedutfordringen med denne oppførselen og hvordan virtuelt minne gjør det mulig. Forklar rollen til sidetabellene.	● 3: The new process needs a copy of the original process' memory. This is necessarily in a different location in physical memory (optional: or swapped out), but both processes use the same addresses and pointers. With virtual memory, it is possible to copy physical memory and copy the old page tables to a new page table, and re-write the page tables to point to the copy. ● 2: The problem is understood, and paging is identified as part of the solution. ● 1: The minimum requires that the student understand that both processes have the same pointers and addresses, and that that is a problem.
3	You are tasked to develop a specialized operating system for a tiny device. Your OS must support the fork() system call with the usual semantics, but the device does not have the hardware to support virtual memory. Can you do it? Explain how or why not.	Du har i oppgave å utvikle et spesialisert operativsystem for en liten utstyrsenhet. OS-et ditt må støtte fork()-systemkallet med den vanlige semantikken, men enheten har ikke maskinvaren til å støtte virtuelt minne. Kan du gjøre det? Forklar hvordan eller hvorfor ikke.	● 3: Yes, but the only possibility is that the processes use the same physical memory. When the OS passes control between parent and child, it must move the entire process to make sure that they always use exactly the same physical addresses. (optional: processes with more virtual memory than physical are impossible; optional: "move" can be replaced with "swap" to indicate movement between disk and physical memory, while "move" can happen inside physical memory) ● 3 (alt): Yes, it is possible to implement an operating system that uses only relative

			addressing. (optional: fork() can rewrite the tiny number of required full addresses.) (Note: XCOFF works like this: AIX, option on MacOS 10.0, BeOS) ● 2: Yes. Followed by something that might work but is highly unreasonable for this context. Like, “Yes. Implement a runtime like the Java Virtual Machine and translate every memory access in software.” ● 1: Yes. nothing else

Topic OS2: lagring

	Question in English	**Spørsmål på norsk**	**Retteveiledning**
	Hard disks (spinning disks) are unfashionable but still preferred wherever large amounts of data are written very often because storage cells of SSDs have a limited number of write cycles. So you can say that hard disks are today only used in situations where you need to know very well how they work.	Harddisker (roterende disker) er umoderne, men fortsatt foretrukket der store mengder data skrives veldig ofte fordi lagringsceller til SSD-er har et begrenset antall skrivesykluser. Så du kan si at harddisker i dag kun brukes i situasjoner hvor du trenger å vite veldig godt hvordan de fungerer.	
1	On a hard disk without Zoning, the most accessed data should be stored very close to the middle track, whereas this data should be stored on tracks located a bit further to the outside of the disk for a disk with Zoning. Explain why.	På en harddisk uten *Zoning* bør de oftest brukte dataene lagres svært nær midtsporet, mens disse dataene bør lagres på spor som ligger litt lenger ut på disken for en disk med *Zoning*. Forklar hvorfor.	● 3: When the disk gets more and more full, data will be stored on all tracks. For unzoned disks, all tracks will hold the same amount of information, for zoned disks, the outer tracks hold more. Since head movement is the slowest part of disk access, it is best to store the most accessed data in the place with the shortest average distance from the average last

			accessed location. For zoned disks, relatively more access will happen on the outside. ● 2: Similar to the points above, but some details are missing. If “more and more full” is missing, still 3 points. No blatantly false information. ● 1: Very little correct information, but the grader believes that the student has understood the higher data density on the outside of a zoned disk.
3	Many operating systems are configured to use the same size for pages in memory and blocks on disk (very often 4096 bytes). Sometimes, however, an administrator chooses to create a partition on a disk (along with a filesystem) that uses either smaller or larger block sizes. Explain (for both larger and smaller blocks) what the benefit is, and give an example of an application that can benefit from each.	Mange operativsystemer er konfigurert til å bruke samme størrelse for sider i minnet og blokker på disk (svært ofte 4096 byte). Noen ganger velger imidlertid en administrator å lage en partisjon på en disk (sammen med et filsystem) som bruker enten mindre eller større blokkstørrelser. Forklar (for både større og mindre blokker) hva fordelen er, og gi et eksempel på en applikasjon som kan være nyttig for hver av dem.	● 3: Larger is more efficient when nearly all files are large, this increases contiguity and reduces file management overhead. For example good for HD video editing. Smaller reduces external fragmentation. For example good for huge number of tiny logfiles. ● 2: Only 2 of the 4 points above are covered. ● 1: Only 1 of the 4 points above are covered.
3	On a zoned disk, would it be preferable to locate partitions with larger block sizes on the outside of the disk and smaller ones on the inside? Explain your reasoning.	Vil det være å foretrekke å finne partisjoner med større blokkstørrelser på utsiden av disken og mindre på innsiden av en *zoned* disk? Forklar resonnementet ditt.	● 3: No. The overhead is the same either way. This is the officially correct answer. ● 2. Yes, with sensible arguments that are probably related to higher speed of reading in outer zones. ● 1. Just “No” without explanation.

Topic OS3: processes

	Question in English	Spørsmål på norsk	Retteveiledning
1	When the operating system decides to perform a context switch between processes, which steps must the operating system perform?	Når operativsystemet bestemmer seg for å utføre en kontekstbytte mellom prosesser, hvilke trinn må operativsystemet utføre?	● 3: remove process from running state, save registers, save complete page tables, depending on the reason for context switch move process to ready queue or set it to blocked; load top-level page table for new process, low registers, move process to running queue ● 2: ⅔ of the points above are present ● 1: ½ of the points above are present
2	What are the similarities and differences between a process and a thread from the point of view of the operating system? (The question refers to kernel threads that are relevant for operating systems, not to user-level threads that some web browsers and interpreted languages use.)	Hva er likhetene og forskjellene mellom en prosess og en tråd fra operativsystemets synspunkt? (Spørsmålet refererer til kjernetråder som er relevante for operativsystemer, ikke til tråder på brukernivå som enkelte nettlesere og interpreterte språk bruker.)	● 3: Differences: Heap memory is shared between threads but not processes. (optional: threads share a page table, processes don't, optional: file handles and sockets are shared between threads but not processes). Similarities: they are scheduled alike, each has its own stack, its own register set ● 2: ⅔ of the points above are present ● 1: ½ of the points above are present
3	Which steps must the operating system perform to create a new process (after a fork() system call)?	Hvilke trinn må operativsystemet utføre for å lage en ny prosess (etter et fork()-systemkall)?	● 3: Make a copy of the original PCB (or process in general), but assign a new process ID. The PCB contains identical registers. Create a new page table, allocate all memory resources and copy their content (optional: some pages can be shared either by being read-only or lazy-copy). Either put the process into ready state or move the original process into ready and new one into running. ● 2: 1 piece of information missing. ● 1: 2 pieces of information missing.

Topic OS4: CPU scheduling

	Question in English	Spørsmål på norsk	Retteveiledning
1	In general, what factors or goals are important to take into account when designing a good CPU scheduler? Which properties should such a scheduler have?	Generelt, hvilke faktorer eller mål er viktige å ta med i betraktningen når du designer en god CPU-planlegger? Hvilke egenskaper bør en slik planlegger ha?	• 3: fast, fair, efficient resource utilization (not only CPU), prevent starving, predictible access, small overhead… (optional/bonsu: mention that there are also a lot of other goals, but they aften depend on target system) • 2: half the points above are present • 1: ⅓ of the points above are present
2	Briefly explain the idea behind a priority-based scheduler.	Forklar kort ideen bak en prioritetsbasert planlegger.	• 3: A thread (or process) is assigned a priority (according to some “rule”), and the processes are organized in round-robin queues according to the priority value. The scheduler then selects processes from the highest priority level first, and then moves to lower priorities. A thread with higher priority will always preempt a thread with lower priority as soon as it arrives. The lower priority thread is blocked until the higher priority process is blocked or terminated. (optional/bonus: dynamic priorities) • 2: half the points above are present • 1: ⅓ of the points above are present
3	You have been hired by BMW to lead the process of making the OS for their next modern electrical car. It will contain a lot of sensors (like rain, temperature, light, etc.), cameras, etc, and there are multiple input sources like steering wheel, pedals, touchpad for media centers, fans, …	Du har blitt ansatt av BMW for å lede prosessen med å lage operativsystemet for deres neste moderne elbil. Den vil inneholde mange sensorer (som regn, temperatur, lys osv.), kameraer osv., og det er flere inngangskilder som ratt, pedaler, berøringsflate for mediasentre, vifter, …	• 3: Create a multi-level scheduler with real-time scheduling in the highest priority class, handling things like breaking, steering, and priority scheduling below. Top pri: driving-related sensors and actors. Mid pri: confort-related devices (heating, fan, light). Low pri: entertainment.

	How will you design a CPU scheduler in such a scenario? Explain your choice with the important aspects and goals to consider in order for the car to operate efficiently and safely - while still providing all the services modern cars provide.	Hvordan vil du designe en CPU-planlegger i et slikt scenario? Forklar ditt valg med de viktige aspektene og målene man må vurdere for at bilen skal fungere effektivt og sikkert – samtidig som den yter alle tjenestene moderne biler tilbyr.	● 2: Priority but not real-time, or real-time but not priority. Consider several priority classes. ● 1: Priority but not real-time, or real-time but not priority.

Topic Net 1: Network Models

	Question in English	**Spørsmål på norsk**	**Retteveiledning**
1	Different approaches have been developed and implemented to address the tasks that a computer network must handle. Both the OSI model and the TCP/IP model follow a common approach but differ in their technical details as well as in their practical applicability. Name this approach in question and explain how the OSI model and the TCP/IP model differ.	Ulike tilnærminger har blitt utviklet og implementert for å løse oppgavene som et datanettverk må håndtere. Både OSI-modellen og TCP/IP-modellen følger en felles tilnærming, men er forskjellige når det gjelder tekniske detaljer så vel som praktisk anvendelighet. Navngi den aktuelle tilnærmingen og forklar hvordan OSI-modellen og TCP/IP-modellen er forskjellige.	● 3: All of the three following criteria fulfilled: - Layered approach identified. - Seven vs five layers explained. - Theoretical application (OSI) vs practical application (TCP/IP), alternatively clear-cut vs fuzzy separation of layers, mentioned. ● 2: Two of the above criteria fulfilled. ● 1: One of the above criteria fulfilled. ● 0: None of the above criteria fulfilled.
2	Apart from the approach shared by the OSI model and the TCP/IP model, other quite different approaches exist. Name and explain briefly two of these approaches.	Bortsett fra tilnærmingen som deles av OSI-modellen og TCP/IP-modellen, finnes det andre ganske forskjellige tilnærminger. Navngi og forklar kort to av disse tilnærmingene.	● 3: Two approaches, that is, component-based and recursive or other relevant alternatives, identified. Both adequately explained. ● 2: Two approaches identified but not adequately explained—or—one approach identified and adequately explained. ● 1: One approach identified. ● 0: None of the above criteria fulfilled.

3	Despite its undeniable shortcomings, the TCP/IP model has been a tremendous success for the modern internet. If you could reset the *status quo* and start over, what approach would you choose? Present your arguments.	Til tross for sine ubestridelige mangler, har TCP/IP-modellen vært en enorm suksess for det moderne internett. Hvis du kunne tilbakestille *status quo* og begynne på nytt, hva slags tilnærming ville du valgt? Legg frem argumentene dine.	• 3: One approach identified, for example RINA. Two sound arguments presented, for example heightened security and privacy, more efficient congestion control, improved scalability and addressing, and support for quality of service. • 2: One approach identified. One sound argument presented. • 1: One approach identified. • 0: None of the above criteria fulfilled.

Topic Net 2: IP addressing

	Question in English	**Spørsmål på norsk**	**Retteveiledning**
	Due to the popularity of the internet, IPv4's address space has been all but exhausted. IPv6 seeks to resolve this issue but is still not as ubiquitous as its predecessor—despite being more than 27 years old.	Grunnet populariteten til internett har IPv4s adresserom nesten blitt oppbrukt. IPv6 søker å løse dette problemet, men er fortsatt ikke så allestedsnærværende som forgjengeren – til tross for at den er mer enn 27 år gammel.	
1	Within the context of IPv4, how has address exhaustion been handled? Identify and explain two approaches.	I sammenheng med IPv4, hvordan har adresseuttømming blitt håndtert? Identifiser og forklar to tilnærminger.	• 3: Two approaches, typically CIDR and NAT, identified and adequately explained. Other relevant approaches include for example reclamation and resale of unused IPv4 addresses. • 2: Two approaches identified, but only one of them adequately explained. • 1: Two approaches identified. • 0: None of the above criteria fulfilled.
2	Consider the IPv4 address block `192.168.0.0/30`. How many addresses	Ta i betraktning IPv4-adresseblokken `192.168.0.0/30`. Hvor mange adresser	• 3: Four addresses stated with adequate explanation.

	does this block contain? Explain how you arrived at your answer.	inneholder blokken? Forklar hvordan du kom frem til svaret ditt.	● 2: Wrong number of addresses stated with adequate explanation. ● 1: Four addresses stated with no or inadequate explanation. ● 0: None of the above criteria fulfilled.
3	IPv6's vast address space makes it possible to assign a unique address to every device connected to the internet. How does this scheme affect security? Discuss pros and cons (at least one each).	IPv6s enorme adresseområde gjør det mulig å tildele en unik adresse til hver enhet som er koblet til internett. Hvordan påvirker denne ordningen sikkerhet? Diskuter fordeler og ulemper (minst én hver).	● 3: At least two well-reasoned and opposing arguments presented. For example: - NAT provides a level of security by obfuscating internal IP addresses. This obfuscation is less present in IPv6. - IPv6 addresses are longer and more complex, which can make traditional scanning-based discovery of devices impractical or even impossible. ● 2: At least two well-reasoned, but reconcilable arguments presented. ● 1: One well-reasoned argument. ● 0: None of the above criteria fulfilled.

Topic Net 3: flow control

	Question in English	**Spørsmål på norsk**	**Retteveiledning**
1	Go-Back-N and Selective Repeat are two popular protocols used for reliable data transmission in computer networks. Discuss the impact of network errors and packet loss on the efficiency of Go-Back-N and Selective Repeat protocols.	Go-Back-N og Selective Repeat er to populære protokoller som brukes for pålitelig dataoverføring i datanettverk. Diskuter virkningen av nettverksfeil og pakketap på effektiviteten til Go-Back-N og Selective Repeat-protokoller.	● 3: explain how both work. go-back-n good for burst errors, selective repeat good for random single packet drop ● 2: like above, but incomplete explanation of drop behaviour, e.g. one good both other wrong og missing ● 1: focus on one only, or only explain both methods, or only explain both recovery behaviours

2	Flow control and congestion control are two key mechanisms that work together to optimize network performance. How do flow control and congestion control work together to optimize network performance? Give two examples.	Flytkontroll og metningskontroll er to nøkkelmekanismer som fungerer sammen for å optimalisere nettverksytelsen. Hvordan fungerer flytkontroll og metningskontroll sammen for å optimalisere nettverksytelsen? Gi to eksempler.	● 3: reduce loss due to in-network and end-system capacity limits. Recover dropped or errant packets efficiently without blocking. ● 2: only the loss reduction arguments ● 1: badly described
3	You are working for a company that designs Internet of Things nodes. These IoT nodes typically have limited processing power, memory, and energy resources. The company would like to implement a flow control algorithm for these nodes. Select a flow control algorithm that may be suitable for such scenarios and explain why.	Du jobber for et selskap som designer Internet of Things-noder. Disse IoT-nodene har vanligvis begrenset prosessorkraft, minne og energiressurser. Selskapet ønsker å implementere en flytkontrollalgoritme for disse nodene. Velg en flytkontrollalgoritme som kan være egnet for slike scenarier og forklar hvorfor.	● 3: stop-and-wait. This is very simple to implement and uses minimal buffers, saving memory (and thus, energy). ● 2: other reasonably simple flow control algos ● 1: shows understanding of the task

Topic Net 4: congestion control

	Question in English	**Spørsmål på norsk**	**Retteveiledning**
1	TCP (Transmission Control Protocol) utilizes congestion control mechanisms, including slow start and congestion avoidance, to manage network congestion and optimize performance. How do slow start and congestion avoidance work? Consider TCP NewReno as presented in the course.	TCP (Transmission Control Protocol) bruker metningskontrollmekanismer, inkludert slow start og congestion avoidance, for å håndtere overbelastning i nettverket og optimalisere ytelsen. Hvordan fungerer slow start and congestion avoidance? Vurder TCP NewReno slik det ble presentert i kurset.	● 3: clearly describe both mechanisms, SS: incremental increase and go to CA when reaching ssthresh. CA: linear increase, multiplicative decrease ● 2: mention both mechanisms but missing points in the description ● 1: if only one mechanism is described

2	Congestion control in wireless networks presents unique challenges due to the characteristics of wireless links, which can introduce packet loss and fluctuating channel conditions. Discuss the challenges of congestion control in wireless networks.	Metningskontroll i trådløse nettverk byr på unike utfordringer på grunn av egenskapene til trådløse lenker, som kan introdusere pakketap og varierende kanalforhold. Diskuter utfordringene med metningskontroll i trådløse nettverk.	● 3: clearly discuss the challenge both in terms of loss and bw variations ● 2: focusing on 1 aspect, loss or bw variations, but clear discussion, OR mention both aspects but without clear discussion ● 1: only discussing one aspect but not enough elaboration on that aspect
3	Incorporating congestion information into routing decisions can be advantageous in certain scenarios. Give one example of such a scenario and explain why such information is beneficial. Today, most routing algorithms do not use the congestion information though, what is the challenge of using congestion information for routing?	Å inkludere informasjon om overbelastning i rutebeslutninger kan være fordelaktig i visse scenarier. Gi ett eksempel på et slikt scenario og forklar hvorfor slik informasjon er nyttig. I dag bruker de fleste rutingalgoritmer ikke metningsinformasjonen, hva er utfordringen med å bruke metningsinformasjon for ruting?	● 3: provides one of the following examples with explanation: load balancing, congestion-aware routing, real time traffic optimization, fault resilient. Mention the challenge: to collect and update the congestion information, or fluctuating routing selection. ● 2: Provide the example without explanation and provide the challenge. ● 1: Either the example or challenge is missing.